

Global Climate Change Indicators: Summary Report 2024

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1. Executive Summary

This annual assessment synthesizes peer-reviewed observational records of the Earth's climate system, greenhouse gas concentrations, and renewable energy transitions across 2023.

2. Atmospheric Greenhouse Gas Concentrations

Global mean atmospheric carbon dioxide concentration reached 421.5 ppm in 2023, representing an increase of 2.8 ppm over the previous year. Atmospheric methane concentration averaged 1922.6 ppb across marine boundary layer monitoring stations.

3. Global Surface Temperature Anomaly

The global mean near-surface temperature in 2023 was 1.45 °C above the 1850-1900 pre-industrial baseline, making it the warmest year on observational record since 1850.

4. Polar Cryosphere and Sea Ice Extent

Antarctic sea ice maximum extent reached 16.96 million sq km in September 2023, marking the lowest winter maximum extent recorded since continuous satellite observations began in 1979.

5. Global Clean Energy Transition

Annual additions of renewable energy capacity reached 510 GW globally in 2023, driven primarily by accelerated installations of solar photovoltaics and wind power infrastructure.